

SAMUVEL NATARAJAN

DevOps Engineer | Cloud & Platform Engineering

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PROFESSIONAL SUMMARY

DevOps Engineer with 3.6+ years of hands-on experience building and automating cloud infrastructure across AWS and Azure. Skilled in Terraform, Kubernetes (AKS/EKS), CI/CD with GitHub Actions and Jenkins, GitOps with ArgoCD, Docker, Linux, and Python/Bash automation. Experienced in cloud monitoring, security, cost optimization, and infrastructure operations, with a track record of reducing provisioning and deployment time through automation. AWS Certified Cloud Practitioner with experience supporting cloud-native applications and platform engineering initiatives.

TECHNICAL SKILLS

Cloud Platforms: Microsoft Azure (AKS), AWS (EC2, EKS, ECR, S3, VPC, IAM, ALB, Route53), Oracle Cloud (OCI)

CI/CD & GitOps: GitHub Actions, Jenkins, and ArgoCD end-to-end pipeline design and automated, GitOps-driven deployments

Infrastructure as Code: Terraform (modular design, remote state, multi-environment: dev/staging/prod)

Kubernetes & Containers: Kubernetes (AKS, EKS), Docker, Helm, Horizontal Pod Autoscaling (HPA), resource limits, ECR, DockerHub

Monitoring & Observability: Prometheus, Grafana, Loki, Alertmanager (Slack & email alerting), real-time dashboards

IAM & Security: AWS IAM policies, security-conscious pipeline automation, Trivy (container image scanning), SonarQube (quality gates)

Linux & Scripting: Linux (Ubuntu, Amazon Linux), Windows Server, Bash/Shell scripting, Python, Jenkins Groovy scripting, YAML, REST APIs

Cloud Networking: VPC/VNet concepts, ALB, Route53 (DNS, health checks), Kubernetes Ingress, SSL termination

Collaboration: Git, GitHub, Agile/Scrum, cross-functional team collaboration

Testing (Supporting): Python, Pytest, and Selenium automated suites supporting release pipelines

PROFESSIONAL EXPERIENCE

DevOps Engineer: Majordomo Inc., April 2023 – Present

Chennai, India

- **Cloud infrastructure & IaC:** Architected and automated scalable cloud infrastructure on AWS and Azure using modular Terraform with remote state management across dev, staging, and production environments, reducing infrastructure provisioning time by ~60%.
- **Cloud cost optimization & governance:** Implemented cost-optimization practices across AWS and Azure development environments using Terraform, resource tagging, budget alerts, and automated scheduling of non-production EC2 instances and VMs to reduce unnecessary cloud spend during off-hours.
- **CI/CD pipelines:** Designed and maintained end-to-end CI/CD pipelines using GitHub Actions and Jenkins covering build, test, security scan, and deploy stages, reducing deployment time by ~70%.
- **Kubernetes & containers:** Deployed and operated containerized microservices on Amazon EKS using Docker and Helm charts; implemented Horizontal Pod Autoscaling (HPA) and resource limits to maintain high availability and cluster performance.
- **GitOps:** Implemented a GitOps continuous deployment model using ArgoCD, managing Kubernetes manifests and Helm charts in a dedicated GitOps repository with automated, zero-downtime rollouts on every Git push.
- **Observability:** Built a centralized observability stack with Prometheus, Grafana, and Loki, including real-time dashboards and Alertmanager rules routed to Slack and email, reducing mean time to detect (MTTD) incidents by ~50%.
- **Security:** Enforced security-first DevOps practices by integrating SonarQube quality gates and Trivy container vulnerability scanning into CI pipelines, achieving zero critical vulnerabilities in production releases.
- **Linux & automation:** Managed Linux-based EC2 instances and wrote Bash scripts for infrastructure automation, log management, and operational tasks in production environments.
- **Quality & release support:** Maintained automated test suites (Python, Pytest, Selenium) to support release confidence, reducing manual QA effort across microservices.

- **Collaboration:** Collaborated with development, QA, and product teams in Agile/Scrum sprints to improve release velocity and system reliability.
- **Platform engineering exposure:** Contributed engineering effort to an internal AI/agent tooling platform (Go backend and MCP tool integration) and to testing an LLM-based AI evaluation feature, broadening exposure to backend and AI-adjacent delivery workflows alongside core DevOps work.

PROJECTS

End-to-End CI/CD, GitOps Deployment & Monitoring Platform for Microservices on AWS

- Designed and implemented a production-grade DevOps platform for containerized microservices on AWS, covering automated infrastructure provisioning, CI/CD, GitOps-based zero-downtime deployments, and centralized observability.
- **Infrastructure as Code:** Provisioned EKS, EC2, VPC, subnets, Route53, ALB, and IAM roles/policies using modular Terraform with S3 remote state and DynamoDB state locking across dev/staging/production.
- **Containers & Kubernetes:** Containerized services with Docker, managed versioned images in ECR/DockerHub, and deployed to EKS via Helm with environment-specific overrides, HPA, and resource limits.
- **CI pipeline:** Built pipelines covering checkout, dependency install, unit testing (Pytest), Docker build, Trivy scanning, SonarQube quality gates, and ECR push, with Slack build notifications.
- **GitOps:** Implemented ArgoCD auto-sync from a dedicated GitOps repository to EKS per environment, with branch protection and PR-based promotion for zero-downtime rollouts.
- **Networking:** Configured ALB with Kubernetes Ingress for traffic routing and SSL termination, and managed DNS/health checks via Route53.
- **Monitoring & alerting:** Deployed Prometheus, Loki, and Grafana for metrics, logs, and dashboards; configured Alertmanager rules that cut MTTD by ~50%.
- **Impact:** ~70% faster deployments, ~60% faster infrastructure provisioning, zero critical CVEs in production, and high availability via HPA-driven autoscaling and zero-downtime ArgoCD deployments.

Cloud-Native DevOps Platform — Personal Project | [GitHub](#)

- Built a production-inspired cloud-native microservices platform using Docker, Kubernetes, Terraform, GitHub Actions, Helm, Ansible, and AWS EKS.
- Designed modular Terraform infrastructure for AWS VPC, IAM, and EKS, including public/private subnets, NAT Gateway, and managed node groups.
- Containerized Node.js/Express and Python/FastAPI services using multi-stage, non-root Docker builds with Kubernetes health probes and resource requests/limits.
- Implemented GitHub Actions CI automation for testing, container builds, and security checks using Trivy, Gitleaks, and Hadolint.
- Automated Linux/server configuration with Ansible and created environment-parameterized Kubernetes deployments using Helm.
- Applied cloud-native security practices, including non-root containers, secret scanning, image vulnerability scanning, and protection of credentials/Terraform state from source control.

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services (Issued 2023)

EDUCATION

Bachelor of Computer Applications (BCA) — Tamil Nadu Open University (TNOU)

Pursuing distance education alongside full-time professional experience | Expected 2027